



Department of Computer Science and Engineering
Academic Year 2022 – 2023 (Even Semester)

Degree, Semester & Branch : IV Semester B.E CSE
Course Code & Title : CS3492 & Database Management Systems
Name of the Faculty member : Mr. K.Vignesh Saravanan, AP / CSE

Innovative Practice Description

- **Unit / Topic: Unit IV / Implementation Techniques** / Query Processing Overview
Algorithms for Selection, Sorting and join operations
- **Course Outcome: CO4**
- **Topic Learning Outcome: TLO 11**
- **Activity Chosen: Collaborative Coding**

- **Justification:**

Developing algorithm is an integral part across all various domains. Collaborative learning is the educational approach of using groups to enhance learning through working together. The prime objective of Collaborative learning is to make the students work as a team to solve problems, complete tasks, or learn new concepts and develop the algorithm for the estimation of query in select, sort, join operations. So, the topic is chosen as the group learning.

- **Time Allotted for the Activity: 40 Minutes**

- **Details of the Implementation:**

After completion of the unit / topics this activity can be planned as a team activity, in order to make the students understand the algorithms very clearly and to improve their level of understanding and gain more knowledge in logical problem solving. The students are provided with materials (like worksheet) with partial codes for algorithms and pseudo code. The task of the students is to fill the correct statements and complete the partially filled algorithms and pseudo codes. The course instructor will provide sufficient hints and correct syntax and will discuss all the doubts in the same class session and make them understand and complete the coding. The screenshots of the worksheet completed by the student's team is shown in Figure.1 and Figure.2. The course instructor also provided the correct answers and explained how the algorithm gets executed and the pseudo code in the end of the session.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO7	PO8	PO9	PO10	PO12	PSO1
CO1	3	3	3	1	1	1	1	2	3

(1 – Low

2 – Moderate

3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PO12
	2	2	3	3	3	2
Justification for correlation	Apply basic Knowledge and fundamentals in Query Processing	Identify the need of measuring the query cost and algorithms	Able to design and develop the Select, Sort and Join algorithms	Functional individually in identifying the algorithm and their cost to implement	Communicate / share the ideas with other students in written and oral presentation	Ability to reproduce the contents gathered through self-learning

Images / Screenshot of the practice:

953621104045
 953621104046
 953621104013
 953621104030
 953621104032

Collaborative Coding (developing algorithms)

Measures of Query Cost and Algorithms

Cost is generally measured as total elapsed time for answering query. Measured by taking into account:

- num. of seeks .
- num. of blocks read
- num. of blocks written .

Cost to write a block is _____ cost to read a block.

The number of block transfers from disk and the number of seeks as the cost measures

$\frac{t_T}{t_S}$ - time to transfer one block
 t_S - time for one seek

Cost for b block transfers plus S seeks

random I/O operation would take $b * t_T + S * t_S$ seconds

Selection Operation

- Algorithm A1 (Linear Search). Scan each file block and test all records to see whether they satisfy the selection condition.
 - Cost estimate = b_r block transfers + 1 seek
 - > b_r denotes number of blocks containing records from relation r
 - Cost = $t_S + b_r * t_T$
 - If selection is on a key attribute (sorted), can stop on finding record
 - > cost = $(b_r/2)$ block transfers + 1 seek
 - Average case $t_S + (b_r/2) * t_T$

Linear search can be applied regardless of

- No selection criteria or
- Non-ordering of records in the file, or
- availability of indices

A2 (primary index, clustering index, equality on key). Retrieve a single record that satisfies the corresponding equality condition.

Cost = $(h_i + 1) * (t_T + t_S)$

A3 (primary index, equality on non key) Retrieve multiple records. Records will be on consecutive blocks (h_i - for the index)

- Let b = number of blocks containing matching records
- Cost = $h_i * (t_T + t_S) + t_S + b * t_T$

A4 (secondary index, equality on key)

- Retrieve a single record if the search-key is a candidate key
- Cost = $(h_i + 1) * (t_T + t_S)$
- Retrieve multiple records if search-key is not a key attribute each of n matching records may be on a different block
- Cost = $(h_i + n) * (t_T + t_S)$

- For relations that fits in memory, Standard sorting - quick sort is the best method.
- For relations that don't fit in memory, External sort is a good choice. It is called as **External Sort**

merge

1. number of sorted runs are created. (small size blocks with some records)

Let i be 0 initially.

Repeatedly do the following till the end of the relation:

(a) read M blocks of relation into memory

Scanned by CamScanner

Figure.1 – Screenshots of the students team - Srivadhani Karthiga A, Subana Devi G, Gokila G.

Collaborative Coding (developing algorithms)

Measures of Query Cost and Algorithms

Cost is generally measured as total elapsed time for answering query. Measured by taking into account:

- Number of seeks * average-seek-cost
- Number of blocks read * average-block-read-cost
- Number of blocks written * average-block-write-cost

Cost to write a block is greater than cost to read a block.

The number of block transfers from disk and the number of seeks as the cost measures

$\frac{t_T}{t_S}$ - time to transfer one block

t_S - time for one seek

Cost for b block transfers plus S seeks

$$b * t_T + S * t_S$$

Selection Operation

- Algorithm A1 (Linear Search). Scan each file block and test all records to see whether they satisfy the selection condition.

- Cost estimate = b_r block transfers + 1 seek

➤ b_r denotes number of blocks containing records from relation r

- $Cost = t_S + b_r * t_T$

- If selection is on a key attribute (sorted), can stop on finding record

➤ cost = $(b_r/2)$ block transfers + 1 seek

- Average case $t_S + (b_r/2) * t_T$

Linear search can be applied regardless of

- No selection criteria or
- Non-ordering of records in the file, or
- availability of indices

A2 (primary index, equality on key). Retrieve a single record that satisfies the corresponding equality condition.

$$Cost = (h_i + 1) * (t_T + t_S)$$

A3 (primary index, equality on non-key) Retrieve multiple records.

Records will be on consecutive blocks (h_i - for the index)

- Let b = number of blocks containing matching records

- $Cost = (h_i + b) * (t_T + t_S) * t_T$

A4 (secondary index, equality on key).

- Retrieve a single record if the search-key is a candidate key

- $Cost = (h_i + 1) * (t_T + t_S)$

- Retrieve multiple records if search-key is not a key attribute each of n matching records may be on a different block

- $Cost = (h_i + n) * (t_T + t_S)$

- For relations that fits in memory, quick sort is the best method.
- For relations that don't fit in memory, merge sort is a good choice. It is called as External Sort

1. Create sorted runs. (small size blocks with some records)

Let i be 0 initially.

Repeatedly do the following till the end of the relation:

- (a) Read M blocks of relation into memory

Scanned by CamScanner

Figure.2 – Screenshots of the students team - Km Architha, Haridharshana M U, Abinaya S

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Student felt good, since they learn the concepts as a group.
 - They felt that through such learning, Students actively participate and this activity encouraged the students to share their knowledge with others.

- ❖ ***Benefit of the practice:***

- Students are actively participated in each group.
 - It helps the students are made to work in a team and share their ideas to others and also improve their communication skills.
 - It helps to focus attention and engage students in learning.
 - Through this activity the students can **deeply understand the algorithmic steps and pseudo code** and how to measure the query costs.
 - The **students are made to write the algorithm and pseudo code with their own understanding** mainly the operations like select, join which gave them complete understanding over the algorithm and pseudo code.

- ❖ ***Challenges faced in implementation:***

- Always the toppers only represent the concepts
 - Some of the students finding difficulties in expressing the ideas

References:

- <https://www.valamis.com/hub/collaborative-learning>
- <https://teaching.cornell.edu/teaching-resources/engaging-students/collaborativelearning>
- <https://www.evergreen.edu/sites/default/files/facultydevelopment/docs/WhatisCollaborativeLearning.pdf>